Exercise Exam Statistics for AI SS2021 (Group 1)

Question 1:

> The temperature in June is normally distributed with an expected value of 22 degrees Celsius and a standard deviation of 5 degrees Celsius.
>
> - a) Calculate the probability that on a random day in June the temperature is between 20 and 27 degrees of Celsius. [6 points]
> - b) Calculate the probability that on a random day in June the temperature is at most 22 degrees of Celsius. [6 points]
> - c) Calculate the probability that on a random day in June the temperature is at least 24 degrees of Celsius. [6 points]
> - d) Calculate a symmetrical interval around the excepted value in which the temperature in June lays with a probability of 60%. [6 points]

a) 0.496766 = 49.677 %
b) 0.50 = 50 %
c) 0.344578 = 34.458 %
d) [17.79 ; 26.21]

Question 2:

> In a class of 30 children 10 have blond hair, 4 red hair, and 16 dark hair. At this same time 5 have green eyes, 2 hazel eyes, 10 blue eyes, and 13 grey eyes. One child is drawn from this class at random.
>
> - a) What is the probability that this child has green eyes or blue eyes? [6 points]
> - b) What is the probability that this child has green eyes or red hair? [6 points]
> - c) What is the probability that this child has hazel eyes or not dark hair? [6 points]

a) 50 %
b) 27.778 %
c) 50.222 %

> for a) you can simply multiply the probabilities for the eye colors
> for b) and c) you have to assume, that hair color and eye color are not conditionally dependent
> then you can calculate the result with the rule for joint sets: P(a or b) = P(a) + P(b) - P(a and b) and the AND is simply the product of the probabilities because of the independence

Question 3:

> A continuous random variable X has the density $f(x) = k*x + k*0.5$ for $0 \leq x \leq 6$ (0 else).
>
> - a) Calculate the constant k so that the probability density function is valid. [7 points]
> - b) State the cumulative distribution function F(x). [7 points]
> - c) Calculate the probability of $P(X \geq 4)$. [6 points]

a) k = 1/21
b) $x/42 + x^2/42$
c) 11/21

Question 4:

A metallurgist measures the length of copper rods at various temperatures and observes the following results. Furthermore, it can be assumed that there is a linear relationship between the length of cooper rods and temperature.

Temperature [degrees Celsius]: 20.4 ; 27.3 ; 39.0 ; 42.9 ; 58.3
Length of the Rod [mm]: 461.12 ; 462.41 ; 464.88 ; 465.69 ; 466.00

- a) Calculate an appropriate summary statistic to quantify the linear relationship between the length of cooper rods and temperature. [15 points]
- b) Which length of a copper rod can be excepted at a temperature of 65 degrees Celsius? [15 points]

a)
mean x = 37.58
mean y = 464.02
covariance = 23.5008
variance x = 172.0936
variance y = 3.6902
Bravais-Pearson Correlation Coefficient: 0.93256 (positve linear correlation)

b)
a = 458.888
b = 0.136558245
xi = 65
yi = 458.888 + (0.136558245 * 65) = 466.8762859 [mm]

Question 5:

The probability that a visit to a particular car dealer results in buying a second-hand car or a Japanese car is 45%. It is known that of those coming to this car dealer 25% buy a second-hand car. It is also known that of those coming to this car dealer 30% buy a Japanese car.

What is the probability that a visit leads to buying a second-hand Japanese car? [8 points]

10 %